

Project — Population Rhythms in E-I Networks

Time to finish: 1 day (about 4-6 hours, working alone or in a pair).

What you will do

You are given recordings of an excitatory (E) and inhibitory (I) neural population, at several levels of external drive. Your job is to test whether the population oscillates, find the oscillation frequency, and check how the frequency depends on the drive.

Background (short)

The Wilson-Cowan model describes two coupled populations of neurons:

E and I $\rightarrow$ they excite and inhibit each other.

When the interactions are strong enough, the network oscillates. Different drive strengths give different oscillation strengths.

Data

In `shared/data/`:

- `population_traces.npz` — E and I activity over 10 seconds, for 48 trials.
- `stimulus_protocol.csv` — the external drive `II` on each condition.

Steps

1. Load and look. Open the starter notebook. Load `E_obs`, `I_obs`, and the protocol. Plot one E and one I trace side by side.
2. Compute the power spectrum. For each trial, use `scipy.signal.welch` to compute the power spectrum of the E signal. Plot the spectrum.
3. Find the peak frequency. For each trial, find the frequency with the highest power. This is the oscillation frequency.
4. Peak frequency vs drive. Plot the mean peak frequency for each condition (`II` value). Write 3-4 sentences describing what you see.

What to hand in

- Your filled notebook.

- The example E and I traces plot.
- One power spectrum plot.
- A plot of peak frequency vs external drive I_L .
- A short paragraph (3-4 sentences) explaining how the oscillation depends on the drive.

If you finish early: compute the total oscillation power (sum of the spectrum around the peak) and plot it vs. drive.